

Lab 11. DNS Server

IT Infrastructure Management (W04IST–SI0826G)

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1 Lab Objectives

1. Installing a DNS server.
2. Configuring zones in a DNS server. Learning record types.
3. Configuring zone delegation.
4. The hosts file.
5. DNS and Active Directory.

2 Exercises

In all exercises, replace `<domain-name>` with your own DNS domain name, e.g. `redbird.internal`. Use `internal` as top-level domain.

2.1 Exercise: Zone Configuration

A DNS server stores records in zones. Each zone can contain one or more DNS domains. A zone is named after the top-level domain defined within it. A DNS server can store zone configuration in Active Directory or in a text file (in this exercise we use the second approach). In this exercise you will configure two types of records: A — resolves a name to an IP address, and CNAME — resolves a name to another name (an alias).

1. On DC1, open the DNS console.
2. Create a new forward lookup zone (Forward Lookup Zones → New Zone). Give the domain your own DNS name:
 - Zone Type: Primary zone, **uncheck** Store the zone in AD,
 - Zone name: `<domain-name>`
 - leave all remaining options unchanged.
3. Set the properties of the `<domain-name>` zone:
 - SOA tab: Responsible person: your email address (use “.” instead of “@”), Refresh interval – 2 min, TTL – 10 min,
 - Zone transfers tab: check the option: Only to servers listed on the Name Servers tab,

- Name Servers tab:
 - click Add,
 - in the FQDN field enter **192.168.56.100** and click Resolve, then OK (note: a message about a non-existing zone will appear – ignore it),
 - click Add.
4. In the **<domain-name>** zone, create records of the following types (note: to get a ping response, we use IP addresses of existing hosts):
 - Host (A): **pc200** → IP address: **192.168.56.200**,
 - Alias (CNAME): **www** → name: **pc200.<domain-name>**. (note the dot at the end of the FQDN).
 5. On CL1, verify that the names **pc200.<domain-name>** and **www.<domain-name>** resolve correctly to IP addresses. Use the commands: **ping** and **Test-Connection**. Pay particular attention to what the command **ping www.<domain-name>** displays.
 6. On DC1 and CL1, in a command-line window, run **nslookup** and execute the following commands:

```
server 192.168.56.200
pc200.<domain-name>
www.<domain-name>
ls -d <domain-name>
ls -t A <domain-name>
ls -t CNAME <domain-name>
```

What does each of these commands do?

What differences did you notice? Which zone setting prevents all information from being displayed on one of the machines?

7. On DC1, run Notepad as Administrator, and open the file: **c:\Windows\System32\dns\<domain-name>**. Then find the records displayed in the previous step.
8. On DC1, open the DNS console and the **<domain-name>** zone. Create a **dev** subdomain inside. In the **dev** domain, create A and CNAME records (with any IP address). Test all records using the **ping** command, e.g. **ping d100.dev.<domain-name>**.

View the **<domain-name>.dns** file again. See how the zone stores the names of domains created within it.

2.2 Exercise: Zone Delegation

If we want a subdomain to be hosted on a different DNS server, we need to create a new, separate zone for it. When the subordinate domain (in this exercise **lab.<domain-name>**) is in a different zone than the parent domain (in this exercise **<domain-name>**), we must define a delegation in the parent zone — that is, add a record pointing to the DNS server that hosts the zone containing the subordinate domain.

1. On DC1, create a new Primary DNS zone named **lab.<domain-name>**.
2. Add the following records to it:

- A – `l200` → `192.168.56.200`
 - CNAME – `www` → `l200.lab.<domain-name>.`
1. To create a link between the parent zone (`<domain-name>`) and the child zone (`lab.<domain-name>`) on DC1, in the `<domain-name>` zone, create a delegation by providing:
 - Delegated Domain Name: `lab` (note the automatically filled FQDN field),
 - Name Servers: click Add, enter the machine: `192.168.56.200` and click Resolve.
 2. On CL1, flush the DNS cache using `ipconfig /flushdns`. View the cache after flushing. Repeat the command: `ping l200.lab.<domain-name>`. Observe how this name is stored in the cache.

2.3 Exercise: The `hosts` File

Before the DNS service was developed, the `hosts` file was used for name resolution. It is always checked **before** the DNS server.

1. On CL1, run the command `ping www.<domain-name>`.
2. (CL1) Run Notepad as administrator and open the file:


```
c:\Windows\System32\drivers\etc\hosts
```

 then add the entry: `8.8.8.8 www.<domain-name>`.
3. Run the `ping` command again. Which IP address does the hostname resolve to now? Which entry is used: the one obtained from the DNS subsystem or from the `hosts` file?

Note: After completing the exercise, remove the entry from the `hosts` file.

2.4 Exercise: DNS and Active Directory

1. On DC1, open the DNS console. Navigate to the `_msdcs.contoso.internal` zone. Examine its contents.
2. Pay attention to the records in this domain:
 - `_tcp.dc._msdcs.contoso.internal` — what does it contain?
 - `_tcp.pdc._msdcs.contoso.internal` — what does it contain?